

Mobile Edge Computing based Intelligent Charging Strategy for Electric Vehicles in Cyber Physical Energy System

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Abstract—The development of intelligent transportation systems has promoted the application of wireless technology in vehicular communication networks and mobile services. At the same time, electric vehicles are popular due to their environmental-friendliness and cost-saving characteristics. However, charging of electric vehicles without proper and real-time rules may cause serious traffic congestion and high expense, which results in unnecessary charging stations and wasted power generation with high carbon emission. Therefore, it is crucial to design an efficient charging strategy for electric vehicles without causing road congestion, together with an economic dispatch strategy to offer proper power generation and decarbonization. In this paper, we introduce mobile edge computing architecture to enhance real-time response capabilities through edge device positioning data. This approach allows charging strategies to swiftly adapt to varying road conditions and user demands, effectively mitigating traffic congestion. Regarding the forecasting of load demand for charging stations, this paper employs a deep learning model based on informer and deploys it on edge servers, quickly providing accurate demand predictions for economic dispatching and enhancing the precision of scheduling strategies. With the application of deep reinforcement learning models, the system can formulate efficient charging plans based on real-time user data, improving user satisfaction and quality of services. Numerical results show the efficiency of the obtained strategies for electric vehicles charging and power economic dispatching, and the two-stage modeling approach significantly improves the convergence of the model and the quality of the solution.

Index Terms—Intelligent transportation system, Electric vehicles charging, Mobile edge computing, Economic dispatch, Deep reinforcement learning.

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I. INTRODUCTION

WITH the development of cyber physical energy system, the application of wireless technology in vehicular communication networks and mobile services has become more and more important. Meanwhile, the transportation systems composed of diesel oil or electric power as energy source are becoming increasingly complex. Electric vehicles (EVs) with renewable energy are popular due to their environmental-friendliness [1] and cost-saving characteristics [2]. For the charging process of EVs, to quickly drive to the charging station, the high traffic flow rate of the road and real-time dispatch become an important demand. To meet such demand, technologies of vehicle-to-vehicle (V2V) [3], vehicle-to-infrastructure (V2I) [4], and vehicle-to-grid (V2G) [5] communications are constructed to sense road conditions, support applications, and enable bidirectional energy flow between EVs and the grid. However, as the number of EVs increases, their charging behaviors will inevitably cause traffic congestion [6], particularly during rush hours, reducing the throughput of the transportation system. In addition, the charging behaviors of EVs without reasonable guidance can also cause uneven power consumption of the charging stations, which will affect the grid's prediction of the load demand [7] of charging stations the next day through historical power consumption data, and may result in energy waste and environmental pollution in the power generation process of smart grid. In this background, integrating mobile edge computing technology becomes a key solution to optimize data processing, improve real-time performance, and enhance intelligent management of transportation systems and charging infrastructure.

There are a series of challenges in designing charging strategies in intelligent transportation systems. Firstly, due to the volatility of energy prices and various speeds of EVs in the road network, it is difficult to design a reasonable charging strategy to guide EVs to suitable charging stations, and this strategy can not only reduce road congestion, but also reduce charging expense for EV users. Secondly, in order to reduce environmental pollution and achieve decarbonization goals, it is crucial to design economic dispatch strategies for maximizing the accommodation of renewable energy according to the energy demand of charging stations. Finally, how the system can rapidly respond to real-time changes in environmental conditions and provide real-time dispatching strategies is also

a significant challenge.

There have been studies in the literature to investigate the EV charging strategies [8]–[10]. These studies generally formulate charging optimization problems for EVs under various scenarios, and design effective charging strategies for the proposed models. However, some key factors in EV charging are not fully considered, such as the uncertainty of energy prices, commuting behaviors of users, and road conditions. These factors will make it difficult for EVs to obtain a better charging strategy. For example, a real-time charging method is proposed to meet the charging needs of users [8]. However, it does not consider the road condition, resulting in EVs without being charged in time due to traffic congestion. The study in [9] incorporated the charging time of EVs into their model and proposed a fast charging navigation strategy. However, they did not consider that the demand for EVs may affect road throughput. In [10], the authors proposed an algorithm for supporting mobile edge computing in charging and discharging networks to minimize the charging waiting time for EVs. However, this algorithm did not take into account the impact of energy price fluctuations at charging stations on the charging strategy, resulting in an increased charging cost for EVs. Therefore, it is necessary to design a comprehensive architecture for EV charging in intelligent transportation systems.

To solve the charging problems, deep learning-based algorithms are designed to obtain optimal charging strategies in intelligent transportation systems, e.g. [11]–[13]. However, if we consider the complex continuous charging task scenario, these methods are not straightforwardly applicable. The authors proposed a distributed charging dispatching method [11], which is based on a deep reinforcement learning (DRL) algorithm deep Q-network to obtain a dispatching strategy for maximizing the profits of charging stations. However, the deep Q-network can only handle discrete actions. For the continuous actions for making EV charging decisions, deep Q-network cannot offer an optimal strategy that maximizes the benefits of charging stations. In [12], the authors considered the uncertainty of EV arrival and the intermittency of renewable energy sources. They described the EV charging framework as an independent Markov decision process (MDP) and converted the EV queue into a mapping of the charging demand queue to minimize the average waiting time for EVs. However, the authors did not take into account the impact of changes in the time factor between the EV and the charging station on the charging strategy. The authors in [13] proposed an EV charging navigation strategy based on price incentive, which can mitigate the pressure of heavy traffic congestion. However, due to the uncertainty of EV charging behaviors, it is a time-consuming task to use traditional methods to solve models compared with DRL-based methods. Therefore, DRL-based algorithms with more considerations on realistic scenarios are crucial.

In this paper, we propose an EV charging strategy incorporating mobile edge computing, aimed at considering economic dispatch and achieving a more intelligent transportation system. To investigate the impacts of charging behavior, we formulate the driving time and charging waiting time of

EVs to indirectly reflect the traffic conditions on the roads. Within this framework, EVs can choose a appropriate charging station for charging tasks according to their own needs. To reduce environmental pollution and power generation costs, we introduce the informer model to predict the total power required for charging stations the next day, and perform economic dispatch through mobile edge computing to rationally allocate power resources. At the same time, we analyze the data collected by edge devices and perform algorithm training based on DRL on the edge servers. This enables the system to learn and adapt in real-time to dynamic road conditions and power supply. Based on this knowledge, we design the charging strategy as part of the customer's application, taking into account the user's charging cost and road congestion, to alleviate traffic congestion and improve user satisfaction. The main contributions of this paper are as follows.

- We propose an EV charging model based on mobile edge computing architecture and considers power economic dispatch, which can quickly achieve low EV charging costs under the premise of minimizing power dispatch cost and carbon emission.
- To efficiently solve the proposed model, we propose a two-layer optimization framework. The first layer of power economic dispatching result affects the second layer of EV charging dispatching process. In turn, the EV charging data collected by edge devices will also affect the result of economic dispatching. The two layers mutually influence and constrain each other during the optimization process, enabling system-level collaborative optimization through iterative coordination.
- In the proposed two-layer optimization process, we decompose the highly complex NP-hard problem into two sequential stages and embed DRL into each stage to enhance the optimization process. Specifically, Soft Actor-Critic (SAC) based DRL algorithms are deployed on edge servers to obtain optimal dispatching results for each stage. Additionally, the charging station load is predicted using an Informer-based algorithm. Numerical results demonstrate that the SAC algorithm outperforms traditional solver-based methods and three other state-of-the-art DRL algorithms in terms of real-time performance, reward convergence speed, and model scalability, while significantly improving the convergence and solution quality of the two-stage model.

The remainder of this article is organized as follows. The Section II mainly introduces related work in terms of intelligent transportation systems and smart grid. Section III describes the system model in this paper. Based on the model, the DRL based algorithms are designed in Section IV. The numerical results are presented in Section V, which demonstrates the performance of our algorithms. Finally, Section VI concludes the whole paper.

II. RELATED WORK

1) Charging Strategies in intelligent transportation systems: Recently, there are different EV charging strategies from different perspectives. In [14]–[16], the authors established a

coupling model of the power network and the transportation network and proposed a charging price incentive control scheme to facilitate the recommendation of charging stations to drivers. In [17], [18], the authors used a multi-agent reinforcement learning method to coordinate the operation of electric vehicle users and distribution networks through a cooperative pricing game, thereby reducing charging prices, alleviating traffic congestion, and solving users' mileage anxiety. In [19], the authors proposed a routing and charging logic-based benders decomposition algorithm to solve the routing and charging scheduling problem of electric passenger vehicles under charging station power constraints and reduce the electricity cost. In [20], the authors proposed a user preference-based charging station recommendation scheme, which combines multi-attribute decision-making theory and considers factors such as charging time, waiting time, cost, and nearby facilities to provide personalized charging station recommendations for electric vehicle users. In [21], the author proposed a parameterized deep Q-network algorithm to solve the charging problem of EVs, and introduced a mobile edge computing framework to maximize the coordination of scheduling resources. The authors proposed an optimization framework for EV joint charging dispatching [22], which used a combination of DRL and binary linear programming (BLP) to solve the problem and obtain the optimal dispatching strategy. The authors proposed a model-free method based on safe deep reinforcement learning [23], and the goal is to find a constrained charge and discharge dispatching strategy to make the charging station meet its own charging needs while also minimizing its charging cost. The authors proposed an adaptive control algorithm for charging of plug-in EVs, which aiming at making the charging behavior of EVs not burden the power system under a low voltage environment [24]. The authors proposed a real-time optimal charging dispatching method for plug-in electric vehicles (PEVs) and developed a parameterized aggregated PEV charging model [25], which aims to solve the problem of the huge charging demands caused by the increasing number of PEVs. To meet the explosive charging demand of EVs, the author jointly considers the charging time optimization of EVs, the pairing of EVs and charging stations, and the pricing mechanism of charging stations [26], and then develops a hierarchical game model to obtain the optimal charging time strategy for EVs.

However, these methods do not take into account the impact of time-varying road conditions, which may make the EVs unable to be charged in time.

2) *Power Economic Dispatch for Smart Grid*: At present, various methods for power economic dispatch are proposed under different scenarios. The authors in [27] proposed a low carbon multi-objective economic dispatching model considering economic costs and environmental costs, to reduce the economic costs of the power system during operation and maintain system constraints. The authors in [28] proposed a new security-constrained multiobjective optimal dispatch (SC-MOOD) framework to realize an economic and reliable operation of microgrids. The author in [29] proposed a renewable energy economic dispatch model with safety constraints, aiming to maximize dispatch efficiency and enhance system

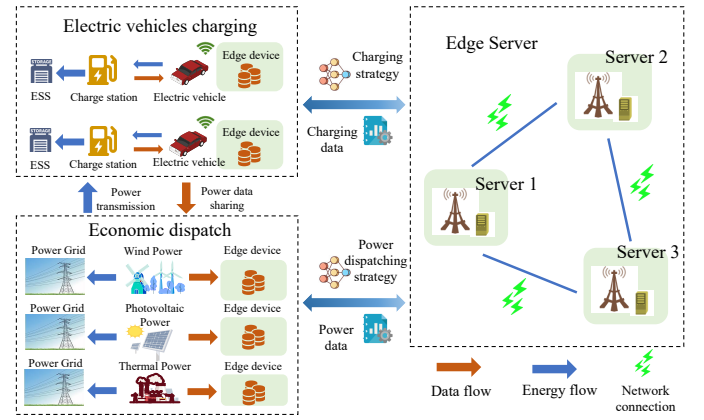


Fig. 1. The architecture of intelligent charging strategy for EVs by considering economic dispatch based on mobile edge computing.

security. A scenario-based robust economic dispatch of virtual power plants in [30], which reduces the operational costs of virtual power plants by using DRL-based algorithms to obtain the dispatch strategy. In [31], the authors proposed a three-stage and multi-timescale online scheduling framework for grid-connected microgrids, which copes with multiple uncertainties by coordinating the energy management of charging stations with the charging scheduling of EVs and achieves efficient scheduling close to the ideal performance. In [32], the authors proposed a new distributed robust optimization method, which uses piecewise linear decision rules and convex difference optimization technology to efficiently solve the real-time power dispatch problem with correlated wind forecast errors and improve dispatch flexibility and economy. In [33], the authors proposed an improved deep reinforcement learning algorithm TCSAC, which solved the multi-objective interval optimization scheduling problem of microgrids under uncertain conditions and effectively improved the economy and stability. In [34], the authors proposed a two-stage data-driven approach to optimize the decentralized dispatch of networked microgrid systems through multi-agent DRL to improve energy balance, cost minimization, and voltage regulation capabilities. In [35], the authors proposed a reinforcement learning-based method to solve the high-dimensional and complex problems in low-carbon economic dispatch by optimizing power flow and carbon capture operations, improving decision-making efficiency and promoting low-carbon operation of the power system.

However, these studies only perform power economic dispatch on the power generation side alone, which may not provide optimal strategies for distributing power to charging stations with dynamic power requirements.

III. SYSTEM MODEL

In this part, we propose an EV charging framework that considers power economic dispatch based on mobile edge computing. Fig. 1 shows the system architecture, which consists of three parts, EV charging, power economic dispatch, and edge server. Firstly, the edge devices collect the historical

load data of the charging stations and analyze it to predict the power required by the charging stations for the next day. It learns the power dispatching strategy through the edge servers and determines the final amount of power dispatched to each charging station. Secondly, at the customer application level, EVs serve as movable edge devices to collect real-time data on charging stations and road traffic, and learn the optimal charging strategy through the edge servers. Finally, the EV selects the appropriate charging station for the charging task based on the strategy. The excess energy generated by the charging stations after a period of time is stored in the energy storage system(ESS).

Fig. 2 depicts an EV charging optimization framework combined with power economic dispatch, which aims to address the problem of EV charging and tackle the key challenge of charging station power dispatch. The framework adopts a two-layer optimization structure. The lower layer is the economic dispatch model of the smart grid, which integrates wind power, photovoltaic, and micro gas turbine power generation. It optimizes the power dispatch plan by utilizing the historical load data of charging stations and reasonably allocates renewable and traditional energy to each charging station to reduce power generation costs and carbon emissions. The upper layer is the EV charging strategy model. EV users select charging stations based on real-time data from the charging stations and road conditions, avoiding charging peak areas, minimizing charging costs, and improving user satisfaction. The two-layer optimization models within the framework are tightly coupled. The power grid indirectly guides EV charging behavior by economically dispatching energy in a reasonable manner, while the actual charging demands and behaviors of EVs dynamically alter the load distribution at charging stations. This information is then fed back to the power grid, driving continuous updates and refinements of dispatching strategies. As the number of iterations increases, the overall accuracy and coordination efficiency of system scheduling are progressively improved, ultimately achieving collaborative optimization between power generation and consumption, enhancing system stability, and reducing environmental impact. The design of the lower-layer economic dispatch model is shown in III-A, and the construction of the upper-layer EV charging strategy model is detailed in III-B.

A. Formulation of Economic Dispatch in the Smart Grid

In this part, energy is generated by wind, photovoltaic and micro gas turbines. The goal of economic dispatch is to minimize the total cost of power generation, including the cost of wind power generation, photovoltaic power generation, micro gas turbine power generation, and the cost of processing the CO_2 released by micro gas turbine.

1) *Wind Power Generation:* To comprehensively investigate the costs of wind power generation, we consider three key parts are considered, including the production cost, the initial depreciation cost, the operation and maintenance cost.

a) *The Production Cost:* The production cost of wind power generation are usually expressed as a quadratic function

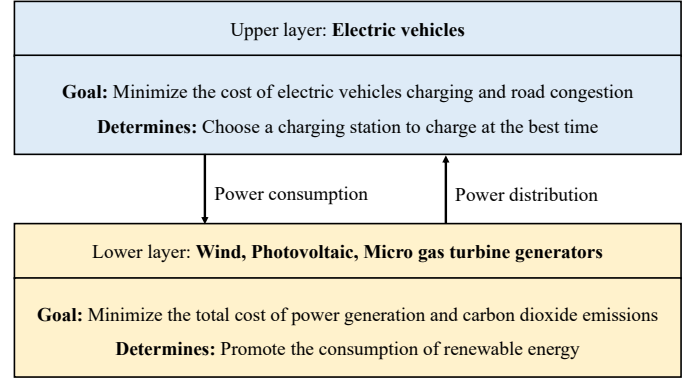


Fig. 2. Two-layer optimization framework of EVs intelligent charging by considering power economic dispatch in the urban transportation system.

of output power [36], which is expressed as

$$C_i^{wpro}(p_i^w(t)) = a \cdot (p_i^w(t))^2 + b \cdot p_i^w(t) + c, \quad (1)$$

where $C_i^{wpro}(p_i^w(t))$ denotes the production cost of the i -th wind generator, and $p_i^w(t)$ denotes output power of the i -th wind generator, and a, b, c denote the coefficients of quadratic function of the production cost.

b) *The Initial Depreciation Cost:* After the wind generators have been used for a certain number of years, the machines themselves will cause an initial depreciation cost [37], which is expressed as

$$C_i^{wdep}(p_i^w(t)) = \frac{C_{wins}}{8760k_w} \cdot \frac{I(1+I)^{n^w}}{(1+I)^{n^w}-1} \cdot p_i^w(t), \quad (2)$$

where $C_i^{wdep}(p_i^w(t))$ denotes the initial depreciation cost of the i -th wind generator, and C_{wins} denotes the installation cost of wind generator unit capacity, and k_w refers to the capacity factor of the wind generator, and n^w is the service life of the wind generator, and I denotes annual rate.

c) *The Operation and Maintenance Cost:* Wind generators need to be maintained regularly during operation, and the operation and maintenance cost [38] will be incurred at this time, expressed as

$$C_i^{wmt}(p_i^w(t)) = G_w \cdot p_i^w(t), \quad (3)$$

where $C_i^{wmt}(p_i^w(t))$ denotes the operation and maintenance cost of the i -th wind generator, and G_w refers to the operation and maintenance cost coefficient of wind generator.

Therefore, the total costs of wind power generation is given by

$$C_i^{wind} = C_i^{wpro}(p_i^w(t)) + C_i^{wdep}(p_i^w(t)) + C_i^{wmt}(p_i^w(t)), \quad (4)$$

where C_i^{wind} is the total power generation cost of the i -th wind generator.

2) *Photovoltaic Power Generation:* Similar to wind power generation, the total costs of photovoltaic power generation are composed of production cost, initial depreciation cost, operation and maintenance cost.

a) *Production Cost*: The production cost of photovoltaic power generation is usually expressed as a quadratic function of output power [36], which is expressed as

$$C_i^{spro}(p_i^s(t)) = a \cdot (p_i^s(t))^2 + b \cdot p_i^s(t) + c, \quad (5)$$

where $C_i^{spro}(p_i^s(t))$ denotes the production cost of the i -th photovoltaic generator, and $p_i^s(t)$ denotes output power of the i -th photovoltaic generator.

b) *Initial Depreciation Cost*: After a certain service period of the photovoltaic generator, there will be initial depreciation cost [37], expressed as

$$C_i^{sdep}(p_i^s(t)) = \frac{C_{sins}}{8760k_s} \cdot \frac{I(1+I)^{n^s}}{(1+I)^{n^s}-1} \cdot p_i^s(t), \quad (6)$$

where $C_i^{sdep}(p_i^s(t))$ denotes the initial depreciation cost of the i -th photovoltaic generator, and C_{sins} denotes the installation cost of photovoltaic generator unit capacity, and k_s refers to the capacity factor of the photovoltaic generator, and n^s is the service life of the photovoltaic generator.

c) *Operation and Maintenance Cost*: The photovoltaic generator also needs operation and maintenance cost [38], expressed as

$$C_i^{smt}(p_i^s(t)) = G_s \cdot p_i^s(t), \quad (7)$$

where $C_i^{smt}(p_i^s(t))$ denotes the operation and maintenance cost of the i -th photovoltaic generator, and G_s refers to the operation and maintenance cost coefficient of photovoltaic generator.

Therefore, the total costs of photovoltaic power generation is given by

$$C_i^{solar} = C_i^{spro}(p_i^s(t)) + C_i^{sdep}(p_i^s(t)) + C_i^{smt}(p_i^s(t)), \quad (8)$$

where C_i^{solar} is the total power generation cost of the i -th photovoltaic generator.

3) *Micro Gas Turbine Power Generation*: Due to the instability of renewable energy generation, micro gas turbine power generation is indispensable, which brings costs from production cost, start and stop cost, operation and maintenance cost, and initial depreciation cost.

a) *Production Cost*: The production cost of micro gas turbine power generation is usually expressed as a quadratic function of output power [36], expressed as

$$C_i^{mpro}(p_i^m(t)) = x_{i,t} \cdot [a \cdot (p_i^m(t))^2 + b \cdot p_i^m(t) + c], \quad (9)$$

where $C_i^{mpro}(p_i^m(t))$ denotes the production cost of the i -th micro gas turbine, and $x_{i,t}$ refers to the running state of the micro gas turbine, with $x = 0$ denoting off and $x = 1$ denoting on, and $p_i^m(t)$ denotes the output power of the i -th micro gas turbine.

b) *Start and Stop Cost*: There will be a cost when the micro gas turbine generators are switched from on to off or vice versa, expressed as

$$C_i^{msc}(t) = f_{msc} \cdot x_{i,t}, \quad (10)$$

where $C_i^{msc}(t)$ denotes the start and stop cost of the i -th micro gas turbine, and f_{msc} refers to the cost required for the start and stop of the micro gas turbine.

c) *Initial Depreciation Cost*: The initial depreciation cost of micro gas turbine power generation is defined as [37]

$$C_i^{mdep}(p_i^m(t)) = \frac{C_{mins}}{8760k_m} \cdot \frac{I(1+I)^{n^m}}{(1+I)^{n^m}-1} \cdot p_i^m(t), \quad (11)$$

where $C_i^{mdep}(p_i^m(t))$ denotes the initial depreciation cost of the i -th micro gas turbine, and C_{mins} denotes the installation cost of the micro gas turbine unit capacity, with k_m referring to the capacity factor of the micro gas turbine, and n^m the service life of the micro gas turbine.

d) *Operation and Maintenance Cost*: The operation and maintenance cost [38] of a micro gas turbine is defined as

$$C_i^{mmt}(p_i^m(t)) = G_m \cdot p_i^m(t), \quad (12)$$

where $C_i^{mmt}(p_i^m(t))$ denotes the operation and maintenance cost of the i -th micro gas turbine, and G_m is the operating and maintenance cost coefficient of the micro gas turbine.

Therefore, the total costs of micro gas turbine power generation is given by

$$C_i^{gas} = C_i^{mpro}(p_i^m(t)) + C_i^{msc}(t) + C_i^{mdep}(p_i^m(t)) + C_i^{mmt}(p_i^m(t)), \quad (13)$$

where C_i^{gas} is the total power generation cost of the i -th micro gas turbine generator.

In addition, micro gas turbine power generation will cause certain amount of pollution to the environment such as CO_2 emission, expressed as

$$F_i^{CO_2}(p_i^m(t)) = x_{i,t} \cdot [\alpha_1 \cdot (p_i^m(t))^2 + \beta_1 \cdot p_i^m(t) + \gamma_1], \quad (14)$$

where $F_i^{CO_2}(p_i^m(t))$ denotes the amount of CO_2 produced by the i -th micro gas turbine, and $\alpha_1, \beta_1, \gamma_1$ respectively denote the pollution emission coefficients of CO_2 .

Therefore, the costs of processing the CO_2 released by the micro gas turbine is given by

$$C_i^{CO_2} = \varphi \cdot F_i^{CO_2}(p_i^m(t)), \quad (15)$$

where $C_i^{CO_2}$ denotes the cost of processing the CO_2 released by the i -th micro gas turbine. The φ denotes the cost conversion coefficient, and its purpose is to convert CO_2 emissions into costs.

4) *Constraints for Economic Dispatch*: The economic dispatch of the smart grid needs to meet various constraints, including output power constraint, power balance constraint and micro gas turbine running state constraint. The specific information is as follows.

a) *Output Power Constraints*: During the power generation process, the generating power of the units cannot exceed its upper and lower limits, expressed as

$$0 \leq p_{i,t}^w \leq \overline{p_{i,t}^w}, \quad (16)$$

$$0 \leq p_{i,t}^s \leq \overline{p_{i,t}^s}, \quad (17)$$

$$\underline{p_{i,t}^m} \leq p_{i,t}^m \leq \overline{p_{i,t}^m}, \quad (18)$$

where $\overline{p_{i,t}^w}$, $\overline{p_{i,t}^s}$ and $\overline{p_{i,t}^m}$ respectively denote the upper limit of output power of wind generators, photovoltaic generators and micro gas turbine, and $\underline{p_{i,t}^m}$ denotes the lower limit of output power of micro gas turbine.

b) *Power Balance Constraint*: The power balance constraint is a basic requirement for the generators, expressed as

$$\sum_{i=1}^L (p_{i,t}^w + p_{i,t}^s + p_{i,t}^m) = \sum_{j=1}^N (p_{j,t}^{load} + p_{j,t}^{rem}), \quad (19)$$

where L is the number of generators, and N denotes the number of charging stations. The $p_{i,t}^w$, $p_{i,t}^s$ and $p_{i,t}^m$ respectively denote the power output of the i -th wind generator, photovoltaic generator, and micro gas turbine, $p_{j,t}^{load}$ and $p_{j,t}^{rem}$ respectively denote power load of the j -th charging station and the remaining power of the j -th charging station transferred to the ESS. In this paper, the strategy of prioritizing the use of stored energy from the ESS is adopted. In equation (19), $p_{j,t}^{rem}$ can take both positive and negative values. Specifically, when $p_{j,t}^{rem} > 0$, it indicates that the ESS is being charged, meaning that excess power from the charging station is being transferred to the ESS for storage. On the other hand, when $p_{j,t}^{rem} < 0$, it indicates that the ESS is discharging, providing stored power to meet the charging station's demand.

c) *Micro Gas Turbine Running State Constraint*: In the process of micro gas turbine power generation, a generator can only be started or stopped at a certain moment, expressed as

$$\sum_{i=1}^L x_{i,t} \leq L, \quad x_{i,t} \in \{0, 1\}. \quad (20)$$

Based on the description above, the goal of power economic dispatch is to minimize the total cost of generators while considering the output and state constraints of the power generation units. Therefore, the objective function for determining the optimal economic dispatch strategy in the first stage can be defined as follows

$$\min_{\mathbf{P}, \mathbf{x}} F_{ed} = \sum_{i=1}^L \sum_{t=0}^{T-1} \left[K_1(C_i^{wind}) + K_2(C_i^{solar}) + K_3(C_i^{gas}) + K_4(C_i^{CO_2}) \right] \quad (21)$$

s.t. (16) – (20),

where $\mathbf{P}$ denotes the generating power of the units, which includes $p_{i,t}^w$, $p_{i,t}^s$ and $p_{i,t}^m$. The $\mathbf{x}$ denotes the running state of the micro gas turbine, which is the same as $x_{i,t}$. The T denotes time slots, and K_1, K_2, K_3, K_4 are the weights for importance of each components, which can be normalized to satisfy $\sum_{i=1}^4 K_i = 1$.

B. Formulation of EV Charging Strategy Problem in intelligent transportation systems

Based on the economic dispatch of the smart grid, each charging station will accordingly receive a different proportion of electricity composed of renewable energy and traditional energy. In intelligent transportation systems, the goal is to minimize users charging cost and road congestion while meeting the charging needs of EVs.

1) *Charging Cost*: The charging cost of an EV can be expressed as

$$C_k^{chg} = \sum_{j=1}^N \sum_{t=t_a}^{t_d} \Delta E_k \cdot p_{j,t}^{ch} \cdot y_{j,k}, \quad (22)$$

where C_k^{chg} denotes the charging costs of the k -th EV, with t_a and t_d denote the start and end of the charging period respectively, and ΔE_k denotes the charge capacity of the k -th EV, and $p_{j,t}^{ch}$ denotes the dynamic energy price of the j -th charging stations at time t , and the energy price here is determined by the third-party operator of the charging stations. There is a slight difference in the hourly energy price of each charging station, which is used to promote EVs to choose charging stations with low energy prices for charging. $y_{j,k}$ denotes that the k -th EV selects the j -th charging station for charging.

2) *Driving Time*: We define the driving time of an EV by the distance and speed of the EV from the charging station, expressed as

$$T_k^d = \sum_{j=1}^N \frac{d_{jk}}{v_{jk}^t} \cdot y_{j,k}, \quad (23)$$

where T_k^d is the driving time of the k -th EV, and d_{jk} denotes the distance between the k -th EV and the j -th charging station, and v_{jk}^t denotes the average speed of the road that the k -th EV travels to the j -th charging stations at time t . Here, d_{jk} and v_{jk}^t are obtained in real-time from third-party interfaces and traffic management departments, respectively.

3) *Waiting Time for Charging*: We consider the case when each charging station is installed on both sides of the road. After an EV reaches the charging station, if there is another EV in front of it, it has to wait for charging. The waiting queue of EVs will affect the throughput of the road, and the waiting time is given by

$$T_k^w = \sum_{j=1}^N t_j^w \cdot y_{j,k}, \quad (24)$$

where T_k^w denotes the waiting time of the k -th EV, t_j^w refers to waiting time required for the EV to select the j -th charging station, which is obtained in real-time from the third-party charging station operation platform.

4) *Road Congestion*: A road can be congested when an EV is driving to a charging station or waiting for charging, due to many EVs to charge. Here we indirectly reflect the degree of road congestion [39] through the driving time and charging waiting time of the k -th EV, which is represented by D_k^{cwd} . The value of driving time and charging waiting time is lower, it means the road traffic flow is greater, and the road congestion is smaller.

$$D_k^{cwd} = \alpha_2 \cdot \frac{T_k^d}{t_0} + \beta_2 \cdot \frac{T_k^w}{t_1}, \quad (25)$$

where $D_k^{cwd} \in [0, 1]$ denotes the congestion degree of the road caused by the k -th EV. The α_2 and β_2 are weight coefficients respectively, which are used to normalize the sum of two calculated values of Eq. (25). The fixed value t_0 denotes time

required for an EV to travel to the charging station at the maximum allowed speed. The fixed value t_1 denotes the maximum tolerable waiting time of EVs. A larger value of D_k^{cud} reflects a higher degree of congestion on the road.

5) *Constraints for EV Charging*: The EV charging needs to meet various constraints, including battery charging constraint, battery power constraint, battery remaining capacity constraint and charging stations selection constraint. The specific information is as follows.

a) *Constraint of Battery Charging*: The charging capacity of the k -th EV cannot exceed the remaining capacity of the charging station, expressed as

$$0 \leq \Delta E_k \leq E_{j,t}^{rem}. \quad (26)$$

b) *Constraint of Battery Power*: The battery capacity of the k -th EV at time t cannot exceed the maximum capacity of the battery at time t , expressed as

$$0 \leq e_{k,t} \leq E_{k,t}^{max}. \quad (27)$$

c) *Constraint of Battery Maximum Charging Power*: The battery charging power of the k th EV at time t cannot exceed its rated power, expressed as

$$0 \leq \rho_{k,t} \leq \mathbb{P}_{k,t}^{max}. \quad (28)$$

d) *Constraint of Battery Remaining Capacity*: The remaining battery capacity of the k -th EV should be enough to reach the charging station, expressed as

$$e_{k,t}^{rem} \geq \sum_{j=1}^N e_{jk}^{min} \cdot y_{j,k}, \quad (29)$$

where $e_{k,t}^{rem}$ denotes the remaining energy of the k -th EV at time t , and e_{jk}^{min} denotes the minimum energy required when the k -th EV travels to the j -th charging station.

e) *Constraint of Charging Stations Selection*: An EV can only choose one charging station. This is expressed as

$$\sum_{j=1}^N y_{j,k} = 1, \quad y_{j,k} \in \{0, 1\}. \quad (30)$$

Based on the description above, the goal of EVs charging dispatch is to minimize users charging cost and road congestion while considering the multiple constraints of EVs in the charging process. Therefore, the objective function for determining the optimal EV charging strategy in the second stage can be defined as follows

$$\min_{\Delta E, \mathbf{y}} F_{ev} = \sum_{k=1}^M \left[K_5 (C_k^{chg}) + K_6 (\vartheta \cdot D_k^{cud}) \right] \quad (31)$$

s.t. (26) – (30),

where ΔE denotes the charge capacity of EV, and $\mathbf{y}$ denotes the EV selects the charging station for charging, which is the same as $y_{j,k}$. The M is the number of EVs, with K_5 , K_6 are the weights for importance of each components, and adjust them to satisfy $\sum_{i=5}^6 K_i = 1$. The ϑ denotes the cost conversion coefficient, and its purpose is to convert road congestion into cost.

Because of the integer constraints (20) and (30), the problems of economic dispatch and EV charging are both NP-hard [40].

IV. PROPOSED METHOD

In this section, we first predict the load demands of the charging stations in the next 24 hours based on the informer model in the edge server for economic dispatching. Secondly, to efficiently solve the NP-hard problems, we propose two DRL algorithms based on SAC [41] to solve the two optimization problems.

A. Consumed Power Prediction Based on the Informer Model

With the popularity of EVs, charging stations are an important part of EV charging facilities, and their power load demand predicting is crucial to the operation and planning of power systems. To achieve accurate prediction of the power load of charging stations better and faster, this paper uses edge devices to collect load data of charging stations, and deploys a method based on the Informer model in the edge server to predict the load demand for the next day.

As a deep learning model, the informer model [42] has many advantages in predicting the power load demand of charging stations in the next day. Firstly, charging station load data usually contains a large amount of time series information, such as hourly or minute-level load data. As a sequence-to-sequence model, the informer model can capture the temporal relationship in time-series data and take into account the influence of historical data when making predictions. Secondly, the informer model adopts a self-attention mechanism, which can automatically learn the long-term dependencies in the data. Therefore, the informer model can better deal with the sequence dependencies in the charging station load data. In addition, the informer model also adopts a multi-layer encoder and decoder structure, which can perform multi-layer abstraction and reconstruction of the input sequence, enabling better capture the feature information in the data. Finally, the informer model has flexible model parameter settings and training methods, which can be customized according to the characteristics and needs of charging station data, so as to obtain more accurate prediction results. In summary, based on the characteristics of the charging station load data, the informer model is a suitable choice for the next day's power load demand prediction of the charging station.

1) The Input of Informer Model:

Fig. 3 shows the overall framework of the informer model. The input of the model consists of two parts, namely $\mathbf{X}_{feed_en}$ and $\mathbf{X}_{feed_de}$. $\mathbf{X}_{feed_en}$ is the input of the encoder, which is usually a sequence of historical observations. $\mathbf{X}_{feed_de}$ is the input of the decoder, which consists of $\mathbf{X}_{token}$ and $\mathbf{X}_0$. And $\mathbf{X}_{token}$ represents a sequence of known length before the start of the predicted target sequence. $\mathbf{X}_0$ represents the placeholder of the target sequence.

2) Self-attention Mechanism of Informer Model:

Firstly, the input form of the traditional self-attention mechanism is (Query, Key), and then scaled dot-product, namely

$$Attention(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = softmax\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V}, \quad (32)$$

where $\mathbf{Q}$ is the query vector, $\mathbf{K}$ is the key vector, $\mathbf{V}$ is the value vector, d_k is the dimension of the projection matrix, $\mathcal{T}$

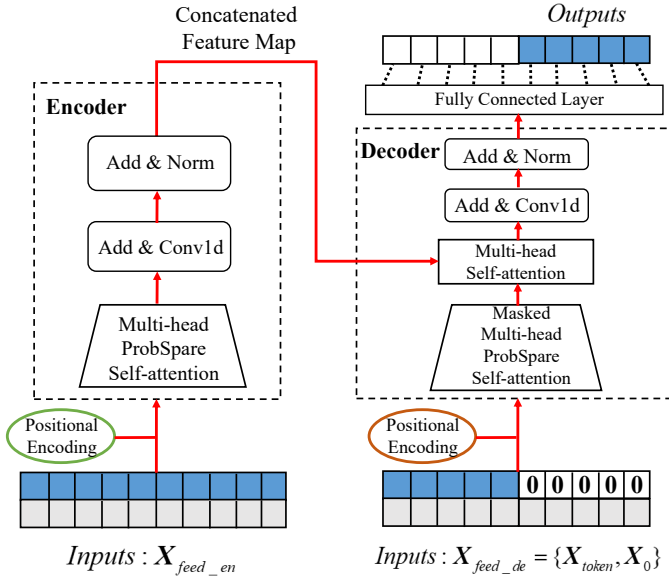


Fig. 3. The overall framework of informer.

represents the transpose operation of the matrix, and $\sqrt{d_k}$ is used to scale attention weights to better control the size of gradients. The probabilistic form of the attention coefficient of the l -th Query is

$$Attention(\mathbf{q}_l, \mathbf{K}, \mathbf{V}) = \sum_g \frac{k(\mathbf{q}_l, \mathbf{k}_g)}{\sum_h k(\mathbf{q}_l, \mathbf{k}_h)} \mathbf{v}_g = \mathbb{E}_{p(\mathbf{k}_g|\mathbf{q}_l)}[\mathbf{v}_g], \quad (33)$$

where $p(\mathbf{k}_g|\mathbf{q}_l) = \frac{k(\mathbf{q}_l, \mathbf{k}_g)}{\sum_h k(\mathbf{q}_l, \mathbf{k}_h)}$, and $k(\mathbf{q}_l, \mathbf{k}_g)$ selects the asymmetric exponential kernel $\exp(\frac{\mathbf{q}_l \mathbf{k}_g^\top}{\sqrt{d_k}})$.

The self-attention mechanism uses the dot-product operation of quadratic time complexity to calculate the above probability $p(\mathbf{q}_l, \mathbf{k}_g)$, and the calculation requires $\mathcal{O}(L_Q, L_K)$ space complexity, where L_Q, L_K respectively denote the input length of query and key, so this is the main obstacle to improving the predictive ability. In addition, in probability research, the probability distribution of self-attention has potential sparsity.

To measure the sparsity of query, Kullback-Leibler divergence is used [43]. The sparsity evaluation formula of the l -th query is

$$M(\mathbf{q}_l, \mathbf{K}) = \ln \sum_{g=1}^{L_K} \exp(\frac{\mathbf{q}_l \mathbf{k}_g^\top}{\sqrt{d_k}}) - \frac{1}{L_K} \sum_{g=1}^{L_K} \frac{\mathbf{q}_l \mathbf{k}_g^\top}{\sqrt{d_k}}, \quad (34)$$

where the first item in (34) is a Log-Sum-Exp of $\mathbf{q}_l$ for all keys, and the second item is their arithmetic mean. Based on the above evaluation method, the formula of *ProbSparse* self-attention can be obtained, namely:

$$Attention(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}(\frac{\bar{\mathbf{Q}} \mathbf{K}^\top}{\sqrt{d_k}}) \mathbf{V}, \quad (35)$$

where $\bar{\mathbf{Q}}$ is a sparse matrix with the same size as $\mathbf{q}$, and it contains only top- u queries under sparse evaluation $M(\mathbf{q}, \mathbf{M})$, the size of u is determined by a sampling parameter. This makes *ProbSparse* self-attention only need to operate

on $\mathcal{O}(\ln L_Q)$ dot-product calculation for each query-key, thus replacing the classic self-attention mechanism.

3) Encoder for the Informer Model:

The encoder is a structure set up to extract long-term dependencies of long-sequence inputs. As a result of the *ProbSparse* self-attention mechanism, there is a redundant combination of value $\mathbf{V}$ in the encoder's feature map. Therefore, the distilling operation is used to assign higher weights to dominant features and generate focus self-attention feature maps in the next layer. The process of distilling operation from layer g to layer $g+1$ is as follows:

$$\mathbf{X}_{g+1}^t = \text{MaxPool}(\text{ELU}(\text{Conv1d}([\mathbf{X}_g^t])), \quad (36)$$

where $\mathbf{X}_{g+1}^t$ and $\mathbf{X}_g^t$ denote layer $g+1$ and layer g sequence respectively, and $\mathbf{X}_{g+1}^t$ is generated by $\mathbf{X}_g^t$. *MaxPool*($\cdot$) means maximizing pooling, *Conv1d* means one-dimensional convolution, and *ELU*($\cdot$) is the activation function. The temporal problems proposed by informer uses a self-attention distilling mechanism, which reduces the length of the input sequence by half at each layer of the decoder, resulting in significant savings in memory usage and computation time for the encoder.

4) Decoder for the Informer Model:

Fig. 3 uses a standard decoder structure, which is composed of a stack of two identical multi-head attention layers. However, the generative inference is used to alleviate the slowdown in long predictions. We provide the following vectors to the decoder as

$$\mathbf{X}_{feed_de}^t = \text{Concat}(\mathbf{X}_{token}^t, \mathbf{X}_0^t) \in \mathbb{R}^{(L_{token}+L_y) \times d_{model}}, \quad (37)$$

where *Concat*($\cdot$) is a concatenation function that concatenates $\mathbf{X}_{token}^t$ and $\mathbf{X}_0^t$ into one long sequence. The *ProbSparse* self-attention applies masked multi-head attention in the calculation. The masked multi-head attention mechanism does not make each position notice the next position, thus avoiding auto-regression. A fully connected layer finally obtains the final output, and whether the prediction is a univariate prediction or a multivariate prediction determines its output dimension. This method uses a generative structure in the decoder structure of the encoder, which can generate all the prediction sequences at one time, so that the time of prediction decoding is greatly shortened.

B. Deep Reinforcement Learning Based Algorithm

In Section III, the economic dispatch and electric vehicle charging scheduling problems are NP-hard due to their integer constraints. Although traditional methods such as CPLEX [44] can obtain exact solutions, their substantial computational overhead makes them impractical for real-time dispatching requirements. To address this, this paper proposes an algorithm based on DRL, which adaptively learns optimal strategy through continuous interaction with the environment. This approach not only effectively reduces computational complexity and eliminates the need for uncertainty modeling but also demonstrates strong adaptability to dynamic changes and large-scale optimization problems.

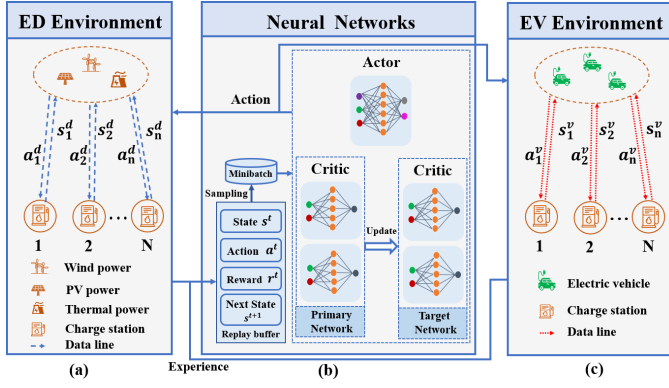


Fig. 4. The structure of the two-stage dispatching algorithm.

The DRL is particularly well-suited for handling complex decision-making problems involving high-dimensional state and action spaces, enabling the discovery of long-term optimal strategies. Moreover, DRL algorithms can be conveniently deployed on edge servers, supporting parallel and distributed inference to meet real-time and scalability requirements. Compared to traditional solver-based methods, the SAC algorithm adopted in this article exhibits faster convergence, higher solution quality, and stronger generalization capabilities in complex environments, significantly improving the overall performance of the scheduling system.

Since DRL is based on MDP, we first transform problems of (21) and (31) as MDP form.

1) Economic Dispatch in Smart Grid:

(a) State: Considering 24 hours a day as a cycle, for the time $t \in (0, 1, 2, \dots, 23)$, the state is defined as

$$s_t^{ED} = \{p_{j,t}^{pred}, p_{j,t}^{load}, p_{j,t}^{rem}\}, \quad j \in N, \quad (38)$$

where $p_{j,t}^{pred}$ denotes the predicted power generation from the prediction model to the j -th charging station.

(b) Action: The agent makes corresponding actions according to the current system states.

$$a_t^{ED} = \{p_{i,j,t}^w, p_{i,j,t}^s, p_{i,j,t}^m, x_{i,t}\}, \quad i \in L, j \in N, \quad (39)$$

where $p_{i,j,t}^w$, $p_{i,j,t}^s$ and $p_{i,j,t}^m$ denote the actual power generated by i -th wind, i -th photovoltaic, and the i -th micro gas turbine to the j -th charging station at time t respectively.

(c) Reward: After the agent makes an action, the system will give the agent an immediate reward, which is also the objective function of the optimization problem (21). The reward function is given by

$$\mathcal{R}_{ED}^{imm}(s_t^{ED}, a_t^{ED}) = -\mathbb{E} \left[\sum_{i=1}^L K_1(C_i^{wind}) + K_2(C_i^{solar}) + K_3(C_i^{gas}) + K_4(C_i^{CO_2}) \right]. \quad (40)$$

After time T , the system will receive total reward

$$\mathcal{R}_{ED} = \max_{\mathbf{P}, \mathbf{x}} \mathbb{E} \left[\sum_{t=0}^{T-1} \zeta^t \mathcal{R}_{ED}^{imm}(s_t^{ED}, a_t^{ED}) \right], \quad (41)$$

where $\zeta \in [0, 1]$ is the discount factor.

2) EVs Charging Dispatching in intelligent transportation systems:

(a) State: Considering the fluctuation of energy prices in 24 hours and the average speed of roads, with $p_j^{ch} = (p_{j,t-23}^{ch}, p_{j,t-22}^{ch}, \dots, p_{j,t}^{ch})$ and $v_j = (v_{j,t-23}^{ch}, v_{j,t-22}^{ch}, \dots, v_{j,t}^{ch})$, The state is divided into three parts, namely EVs, charging stations and road traffic, the details are as follows

$$s_t^{EC} = \{Part_1, Part_2, Part_3\}, \quad (42)$$

where

- $Part_1 = \{e_{k,t}, d_{jk}, e_{jk}^{min}, E_{k,t}^{max}, \rho_{k,t}, \mathbb{P}_{k,t}^{max}\}$: respectively denotes the power of EV, the distance from EV to the charging station, the minimum energy required when EV travels to the charging station, the maximum capacity of the battery for EV, the battery charging power of the EV and the rated power of the EV.
- $Part_2 = \{p_{j,t}^{ch}, E_{j,t}^{rem}, t_j^w\}$: respectively denotes the energy price of the charging station, the remaining power of the charging station and the waiting time required for the charging station.
- $Part_3 = \{v_{jk}^t\}$: denotes the average speed of the road.

(b) Action: According to the current state, the EV, which acts as the agent in this model, determines a proper action based on the observed state information. The action is defined as

$$a_t^{EC} = \{y_{j,k}, \Delta E_k\}, \quad j \in N, k \in M. \quad (43)$$

(c) Reward: When the EV takes an action, the system will give the agent an immediate reward, which is also the objective function of the optimization problem (31),

$$\mathcal{R}_{EC}^{imm}(s_t^{EC}, a_t^{EC}) = -\mathbb{E} \left[\sum_{k=1}^M (K_5(C_k^{chg}) + K_6(\vartheta \cdot D_k^{wd})) \right]. \quad (44)$$

After time T , the system will receive total reward

$$\mathcal{R}_{EC} = \max_{\Delta \mathbf{E}, \mathbf{y}} \mathbb{E} \left[\sum_{t=0}^{T-1} \eta^t \mathcal{R}_{EC}^{imm}(s_t^{EC}, a_t^{EC}) \right], \quad (45)$$

where $\eta \in [0, 1]$ is the discount factor.

C. Two-stage DRL Algorithms in Edge Servers

Due to the real-time requirements of the system are high, in the two-stage scheduling process, we use edge intelligent devices to collect real-time data and deploy the DRL model in the edge server to achieve an efficient scheduling strategy.

In the process of power economic dispatching and EV charging dispatching, SAC based algorithms are designed to obtain the optimal policy, since the state transition of the system suffers continuous changes with high-dimensionality. Besides, the SAC generates actions based on the maximum entropy theory under off-policy, which solves the shortcomings of the high sampling complexity of on-policy reinforcement learning, and improves the fragile convergence of the reinforcement learning.

Based on the framework of SAC, we define soft state value function network $V_\psi(s^t)$, soft Q-function network $Q_\theta(s^t,$

Algorithm 1 EDSAC.

Input: Predicted power generation $p_{j,t}^{pred}$, energy storage capacity $p_{j,t}^{rem}$, load power $p_{j,t}^{load}$.
Output: Energy ratio of each charging station a_t^{ED} .
1: Initialization: policy parameters π_ϕ , soft Q-function parameters Q_{θ_1} , Q_{θ_2} , V-function parameters V_ψ , target networks parameters $\hat{\psi} \leftarrow \psi$, and empty replay buffer $\mathcal{D}_1$
2: **for** each Episode **do**
3: **for** each environment Step **do**
4: Obtain state s_t^{ED} from environment via (38);
5: Input s_t^{ED} into actor network and get action $a_t^{ED} \sim \pi_\phi(\cdot | s_t^{ED})$ via (39);
6: Get immediate reward $\mathcal{R}_{ED}^{imm}(s_t^{ED}, a_t^{ED})$ via (40), next state s_{t+1}^{ED} , and done signal d_{ED} from the environment through a_t^{ED} ;
7: Store the tuple $\langle s_t^{ED}, a_t^{ED}, \mathcal{R}_{ED}^{imm}(s_t^{ED}, a_t^{ED}), s_{t+1}^{ED}, d_{ED} \rangle$ into replay buffer $\mathcal{D}_1$;
8: **end for**
9: **for** each gradient Step **do**
10: Randomly sample a mini-batch of samples from the replay buffer $\mathcal{D}_1$;
11: Compute target values;
12: Update V-function parameters ψ via (46);
13: Update soft Q-function parameters θ_1, θ_2 via (47);
14: Update the policy parameters ϕ via (48);
15: Update target network with:
 $\hat{\psi} \leftarrow \tau\psi + (1 - \tau)\hat{\psi}$;
16: **end for**
17: **end for**

a^t) and policy function network $\pi_\phi(a^t | s^t)$ with deep neural networks, respectively. In addition, a target network of soft state value function $V_{\hat{\psi}}(s^t)$ is implicitly defined in SAC.

To train the soft state value function network parameters, MSE is used to minimize the residual expressed as

$$J_V(\psi) = \mathbb{E}_{s^t \sim \mathcal{D}} \left[\frac{1}{2} (V_\psi(s^t) - \mathbb{E}_{a^t \sim \pi_\phi} [Q_\theta(s^t, a^t) - \log \pi_\phi(a^t | s^t)])^2 \right], \quad (46)$$

with states sampled from a replay buffer $\mathcal{D}$ and actions generated by the current policy.

Similarly, the objective function of the soft Q-function $Q_\theta(s^t, a^t)$ also uses MSE to minimize the soft Bellman residual, which is expressed as

$$J_Q(\theta) = \mathbb{E}_{(s^t, a^t) \sim \mathcal{D}} \left[\frac{1}{2} (Q_\theta(s^t, a^t) - \hat{Q}(s^t, a^t))^2 \right], \quad (47)$$

where $\hat{Q}(s^t, a^t)$ is the estimated value of soft Q objective function.

The objective function of our policy $\pi_\phi(a^t | s^t)$ is to minimize the Kullback-Leibler divergence between the two distributions, which is expressed as

$$J_\pi(\phi) = \mathbb{E}_{s^t \sim \mathcal{D}, a^t \sim \pi_\phi(a^t | s^t)} \left[\log \pi_\phi(f_\phi(\varepsilon^t; s^t) | s^t) - Q_\theta(s^t, f_\phi(\varepsilon^t; s^t)) \right]. \quad (48)$$

TABLE I
PARAMETER SETTINGS.

Variable	Distribution	Range
Charging price(\$/kWh)	$p_{j,t}^{ch} \sim N(m_p, \delta_p^2)$	$0 \leq p_{j,t}^{ch} \leq 5.0$
Average road speed(km/h)	$v_{jk}^t \sim N(m_v, \delta_v^2)$	$0 \leq v_{jk}^t \leq 90$
Distance from EV to charging stations(km)	$d_{jk} \sim U(0.5, 9.0)$	$0.5 \leq d_{jk} \leq 9.0$
Waiting time for EV charging(h)	$t_j^w \sim U(0.05, 0.50)$	$0.05 \leq t_j^w \leq 0.50$
The capacity of battery (Ah)	$e_{k,t} \sim U(10, 120)$	$10 \leq e_{k,t} \leq 120$
The charging power of battery (kW)	$\rho_{k,t} \sim U(0, 8.0)$	$0 \leq \rho_{k,t} \leq 8.0$
Minimum energy required for EV to drive to charging stations(kWh)	$e_{jk}^{min} \sim U(1, 100)$	$1 \leq e_{jk}^{min} \leq 100$
$m_p = (0.65 * 5.0), \delta_p^2 = (0.15 * 5.0)^2$. $m_v = (0.61 * 90), \delta_v^2 = (0.28 * 90)^2$.		

In the first stage, we propose an economic dispatching algorithm based on SAC(EDSAC). Fig. 4(a) shows the agent collects the state from the charging stations, takes actions from the deep neural network, and receives rewards or penalties, which tries to learn an optimal policy to allocate a different proportion of energy to each charging station. Fig. 4(b) shows the entire neural network is composed of the primary critic network, target critic network, the actor network, and the replay buffer. The specific process is shown in Algorithm 1.

The complexity of Algorithm 1 is low, it is mainly determined by 2 soft Q networks with Z_1 fully connected layers, 2 value networks with Z_2 fully connected layers, and 1 policy network with Z_3 fully connected layers. The time complexity can be calculated as [45]

$$2 \times \sum_{z_1=0}^{Z_1} n_{\theta, z_1} \cdot n_{\theta, z_1+1} + 2 \times \sum_{z_2=0}^{Z_2} n_{\psi, z_2} \cdot n_{\psi, z_2+1} + \sum_{z_3=0}^{Z_3} n_{\phi, z_3} \cdot n_{\phi, z_3+1} = \mathcal{O} \left(\sum_{z_1=0}^{Z_1} n_{\theta, z_1} \cdot n_{\theta, z_1+1} + \sum_{z_2=0}^{Z_2} n_{\psi, z_2} \cdot n_{\psi, z_2+1} + \sum_{z_3=0}^{Z_3} n_{\phi, z_3} \cdot n_{\phi, z_3+1} \right), \quad (49)$$

where n_{θ, z_1} , n_{ψ, z_2} and n_{ϕ, z_3} denote the unit number in the z_1 -th soft Q network layer, z_2 -th value network layer and z_3 -th policy network layer respectively.

After the economic dispatch in smart grid, each charging station receives different proportions of renewable energy and traditional energy. In the second stage, we propose an EV charging dispatching algorithm based on SAC(EVSAC). Fig. 4(c) shows the EV in the area is an agent, which collects the state from the charging station, and tries to learn the optimal charging policy. The specific algorithm steps are similar to Algorithm 1, where the input is the output of Algorithm 1, road conditions, and EV power, and the output of the EVSAC is the selection of a certain charging station and how much electricity is charged. The state, action, and reward of EVSAC are in the form of MDP in part IV-B2. As the two-stage DRL algorithm is continuously executed alternately,

TABLE II
PREDICTIVE EVALUATION INDICATORS OF THE THREE MODELS.

Indicators	MSE	RMSE	MAE	MAPE
Models				
Informer	1.936	1.392	1.015	1.962%
LSTM	5.088	2.256	1.592	3.220%
RNN	11.230	3.351	2.239	4.387%

the model's scheduling strategy is also constantly adjusted accordingly until system-level collaborative optimization is achieved.

V. NUMERICAL RESULTS

A. Simulation Setup

In the simulation, we aim to simulate the energy management and electric vehicle charging dispatching problem in an edge computing environment. To better reflect the characteristics of edge computing, we first configured multiple energy generation equipment, including 5 wind generators, 5 photovoltaic generators and 5 micro gas turbines. These power generation devices are distributed in different geographical locations, simulating the diversity and distribution of edge nodes. To simulate real-world conditions, we used real data sets from elia group [46], including photovoltaic power generation, wind power generation and load data, taking into account the volatility of different types of energy sources. For the EVs charging dispatch in the second stage, we arranged multiple charging stations belonging to the same operator and multiple intersecting roads, and fully considered the layout of edge nodes in high-density areas such as urban centers and traffic intersections. As suggested in [47], the charging price and the average road speed are subject to truncated normal distributions with different means and standard deviations, respectively, and the remaining variables are subject to uniform distribution. The specific parameter settings are shown in Table I.

For the EDSAC algorithm, we train the model using three types of networks, soft state value network, soft Q network, policy network, respectively. The structure of soft state value network is (3, 256, 256, 1, $relu(x)$, Adam), and the structure of soft Q network is (11, 256, 256, 1, $relu(x)$, Adam), with the structure of policy network is (3, 256, 256, 8, $\frac{\tanh(x)+1}{2}$, Adam). For the EVSAC algorithm, the structure of soft state value network is (19, 128, 128, 1, $relu(x)$, Adam), and the structure of soft Q network is (24, 128, 128, 1, $relu(x)$, Adam), with the structure of policy network is (19, 128, 128, 5, $\frac{\tanh(x)+1}{2}$, Adam). The size of the replay buffer is set to 10^6 , and the size of a mini-batch sample is 64.

All the simulations are run on a machine with a 4-core i5 CPU, a 12GB RAM and a GTX960M GPU. The version of Python is 3.8, with the version of Pytorch and TensorFlow 1.8 and 2.4 respectively.

B. Electricity Prediction Results of Charging Stations

To reduce the excess energy and environmental pollution, we need to make a prediction of the electricity demand of all

TABLE III
COMPARISON OF TIME TAKEN BY DRL ALGORITHM AND CPLEX IN CALCULATING POWER GENERATION.

Models	EDSAC	PPO	A3C	DDPG	CPLEX
Infer Time(s)	2.972	3.294	3.961	3.186	927.6

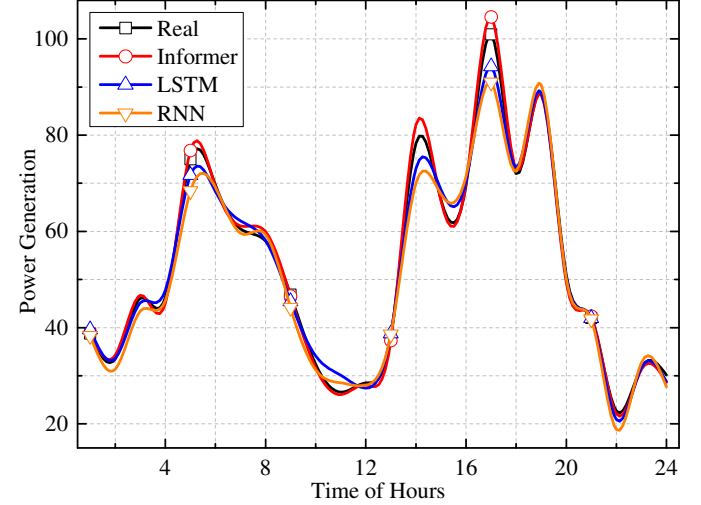


Fig. 5. Total power generation forecast of charging stations in the next 24 hours.

charging stations in the next 24 hours, which offers enough time for decision-making. Table II shows the comparison of the evaluation indicators of the three prediction models for predicting the power data in the next 24 hours. "MSE" denotes mean squared error, which is used to evaluate the degree of change in the data. "RMSE" represents root mean square error, which is a measure of the dispersion of the deviation between predicted power data and actual power data. "MAE" denotes mean absolute error, which reflects the prediction performance of the average magnitude of the model's prediction error. "MAPE" refers to mean absolute percentage error, which reflects the overall average performance of a predictive model. The smaller the value of the above four indicators, the smaller the prediction error of the model.

Fig. 5 depicts the results of the predictive model to predict the total power of the charging stations in the next 24 hours. From Fig. 5, we can see that the model of informer fits the "Real" curve better than the LSTM, RNN models.

C. Economic Dispatch Results

Fig. 6 shows the normalized average cumulative rewards of under different learning rates (lr). When the number of training episodes reaches 4,000, the circular curve and the triangular curve all converge to the same reward value. However, the circular curve converges much faster than the triangular curve. When $lr = 10^{-4}$, although the reward shows an upward trend during the first 1,500 episodes of training, as the number of episodes increases, the average cumulative reward has shown a clear downward trend. On the whole, the EDSAC algorithm shows the best performance when $lr = 10^{-5}$. Therefore, the

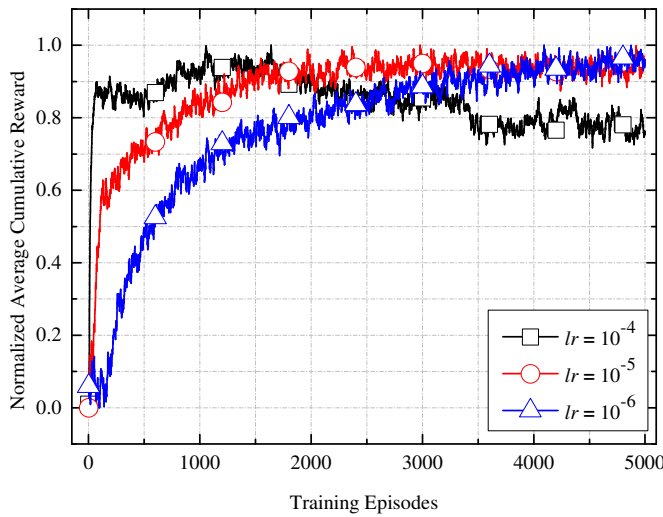


Fig. 6. The convergence process of EDSAC algorithm under different learning rates.

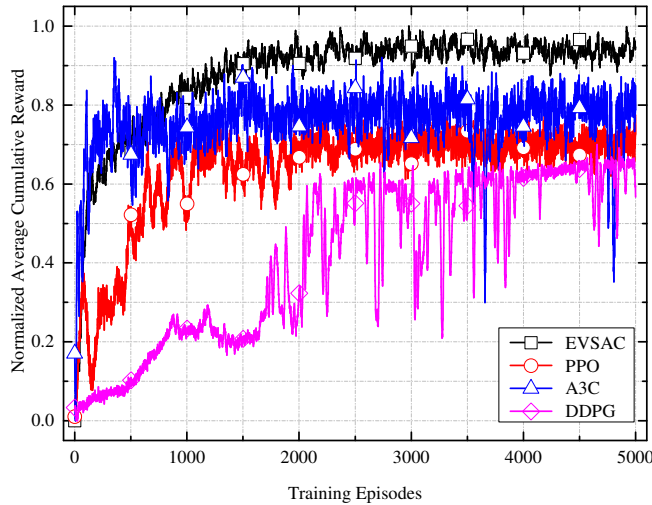


Fig. 7. Comparison of rewards of different DRL algorithms in power economic dispatch optimization when $lr = 10^{-5}$.

different network learning rates have a great impact on the convergence trend of rewards.

Fig. 7 illustrates the reward convergence comparison of different DRL algorithms for solving the power economic dispatch optimization problem at a learning rate (lr) of 10^{-5} . As shown in Fig. 7, the reward value of the EDSAC algorithm exceeds that of the other three DRL algorithms upon convergence, demonstrating that the EDSAC algorithm yields an optimal power dispatch strategy, effectively minimizing the total cost of power generation under multiple constraints.

Table III presents a comparison of computation times for different DRL algorithms and the CPLEX solver in calculating optimal power generation on the same power test dataset. As shown in Table III, the DRL algorithms require less than 3 seconds to compute, whereas the solver demands approximately 300 times more computation time. This significant difference is one of the reasons the solver was not chosen.

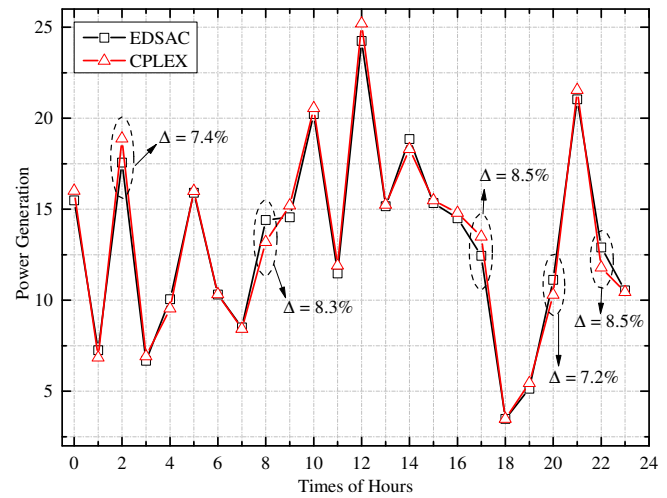


Fig. 8. Comparison of power generation results using DRL and solver CPLEX for power economic dispatch.

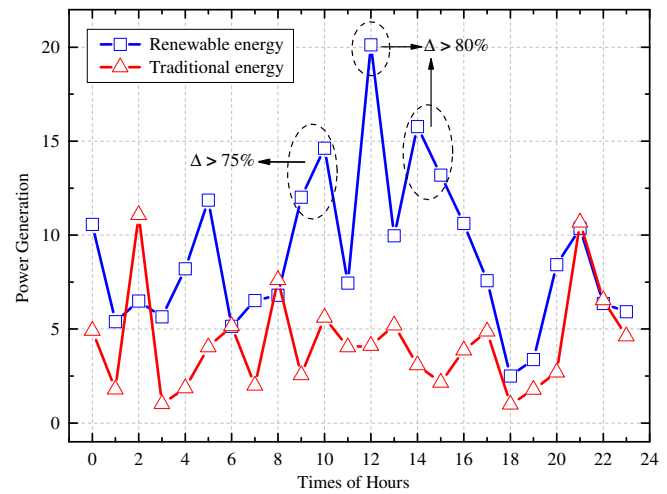


Fig. 9. The ratio of renewable energy to traditional energy for the first charging station in a single day.

Unlike the solver, DRL leverages historical data offline to train a reinforcement learning model, enabling it to quickly provide optimal strategies online when new data becomes available.

Fig. 8 shows the comparison of the power generation results given by the DRL and CPLEX solvers for the power economic dispatch optimization problem. The symbol Δ represents the deviation ratio between the approximate optimal solution given by the EDSAC algorithm and the optimal solution calculated by the CPLEX solver. Since DRL operates in a high-dimensional and dynamic environment, strict mathematical constraints are not always explicitly enforced at every decision step, which may result in potential violations. To address this issue, we integrate constraints into the DRL framework through three complementary mechanisms: action clipping to enforce power output limits, reward penalties to discourage power imbalance, and rule-based filtering to maintain the discrete operational states of generation units. These strategies collectively reduce constraint violations and enhance the feasibility of the resulting dispatch plans. As shown in Fig. 8,

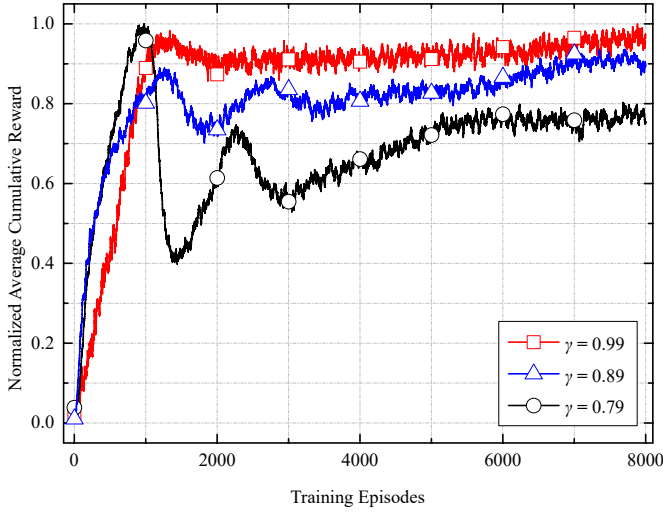


Fig. 10. The convergence process of EVSAC algorithm under different discount rates.

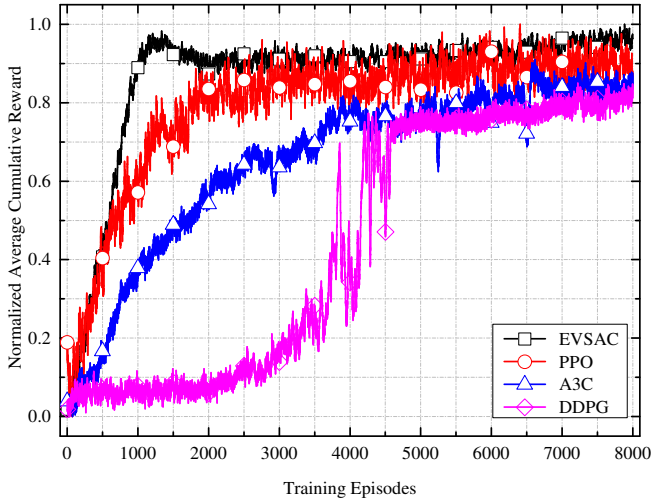


Fig. 11. Comparison of rewards of different DRL algorithms in EVs charging dispatch optimization when $\gamma = 0.99$.

the maximum deviation ratio remains consistently stable at approximately 8.5%. Although there is a slight difference compared to the optimal solution, the use of an ESS can effectively mitigate this deviation. Furthermore, this difference is theoretically justifiable, as DRL is designed to address high-dimensional, dynamic, and uncertain optimization problems. The solving process relies on the agent's strategy learning and interaction with the environment to approximate the optimal solution. Therefore, the observed slight deviations above or below the optimal solution, such as those at 8:00, 20:00, and 22:00, are explainable.

Fig. 9 shows the ratio of renewable energy to traditional energy at the first charging station, where Δ is the utilization rate of renewable energy. It can be seen from Fig. 9 that the square curve is above the triangular curve in most of the 24 hours a day. For example, when 9:00~10:00 in the morning, due to the increase in the demand for charging of EVs, the utilization rate of renewable energy even exceeds 75%. In

TABLE IV
OVERALL TIME DELAY.

(M, N)	(2,5)	(4,5)	(6,5)	(8,5)	(10,5)
Time(ms)	96.32	113.9	132.45	158.6	187.68

TABLE V
COMPARISON OF THE TIME TAKEN BY DRL ALGORITHM AND CPLEX TO FIND THE OPTIMAL CHARGING STATION FOR CHARGING.

Models	EVSAC	PPO	A3C	DDPG	CPLEX
Infer Time(s)	3.444	2.238	2.904	2.196	766.2

TABLE VI
COMPARISON OF DRL AND CPLEX ON EVS CHARGING DISPATCH OPTIMIZATION OBJECTIVES.

Models	Objective	C^{chg}	T^d	T^w	D^{cwl}
	EVSAC	88.427	0.333	0.244	0.291
	PPO	89.631	0.375	0.295	0.343
	A3C	91.775	0.429	0.327	0.385
	DDPG	95.388	0.46	0.366	0.424
	CPLEX	85.044	0.287	0.217	0.256

general, the utilization rate of renewable energy generation within a single day of the first charging station is higher than that of traditional energy generation.

D. EVs Charging Dispatch Results

Table IV presents the overall response delay of the system under different numbers of edge nodes, where M is the number of EV nodes and N is the number of charging station nodes. As the number of nodes increases, the overall delay of the system is also in the hundreds of milliseconds, meeting the real-time requirements of the system.

Fig. 10 shows the normalized average cumulative rewards of the proposed EVSAC algorithm under different discount rates. When the number of episodes is before 1,500, the average cumulative rewards under the three discount rates show a gradually increasing trend. However, when the number of episodes is between 1,500 and 4,000, the reward curves of $\gamma = 0.79$ and $\gamma = 0.89$ decrease significantly. When the number of episodes reaches 8,000, the average cumulative rewards under the three discount rates both reach the convergence state, but the average cumulative reward when $\gamma = 0.99$ is greater than that under the other two discount rates. From the overall effect, when $\gamma = 0.99$, the performance of the proposed EVSAC algorithm is optimal. Therefore, different discount rates have a great impact on the rewards.

Fig. 11 shows the comparison of normalized average cumulative rewards achieved by four different DRL algorithms when addressing the EVs charging dispatching problem in the same environment, with the discount factor $\gamma = 0.99$. As shown in Fig. 11, the proposed EVSAC algorithm outperforms the other three state-of-the-art DRL algorithms in terms of both convergence speed and final reward value.

Table V presents the comparison of computation times required by the different DRL algorithms and the CPLEX

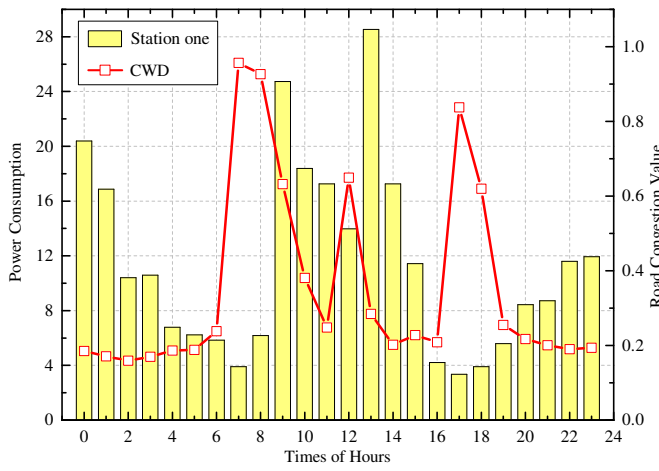


Fig. 12. The relationship between the energy consumption of the first charging station in a single day and the degree of road congestion.

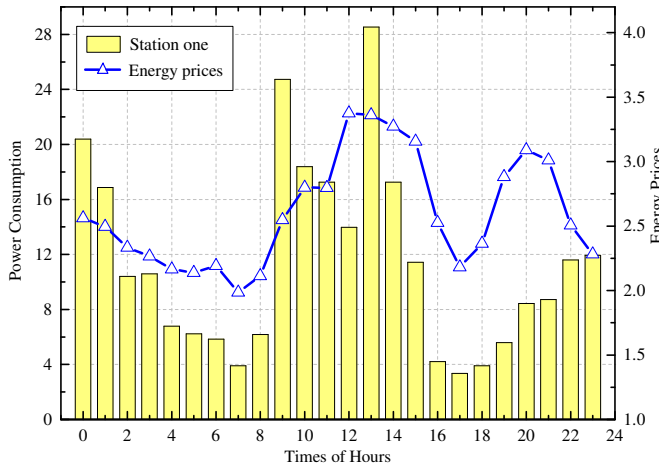


Fig. 13. The relationship between the energy consumption of the first charging station in a single day and the fluctuation of energy prices.

solver to identify the optimal charging station on the same test dataset in the charging environment. As shown in Table V, the four state-of-the-art DRL algorithms are approximately 220 times faster than the CPLEX solver, effectively meeting the real-time requirements of the charging system.

Table VI depicts the comparison of the optimization results for the EVs charging dispatching objectives between the DRL and CPLEX solver methods. As shown in Table VI, the optimization performance of EVSAC is nearly identical to that of CPLEX in terms of four objectives: charging cost, travel time, queue waiting time, and road congestion, which further demonstrates the effectiveness of the DRL algorithm we used.

Figs. 12 and 13 depict the influencing factors of the energy consumption of the first charging station in one day, where the histogram denotes the energy consumption of the first charging station at each time, and the square curve in Fig. 12 denotes the degree of road congestion, the triangular curve in Fig. 13 denotes the price of energy. From Figs. 12 and 13, we can see that during the 6:00~8:00, the road congestion gradually

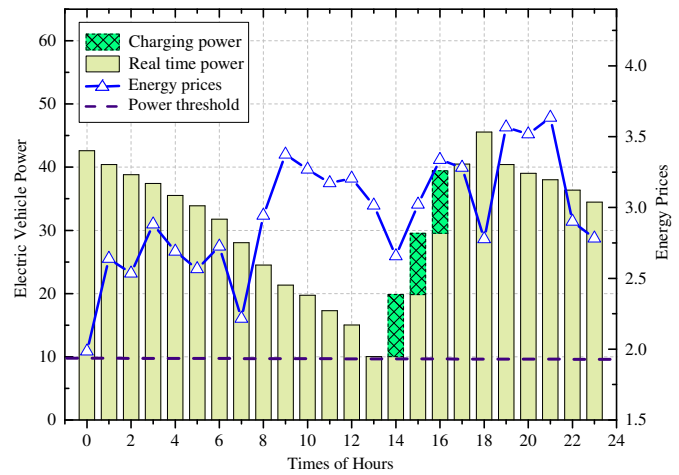


Fig. 14. The relationship between the charging behavior of electric vehicle in a single day and the fluctuation of energy prices.

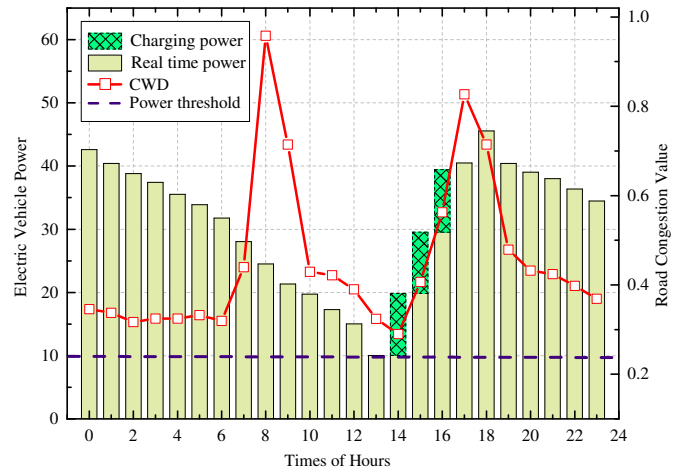


Fig. 15. The relationship between the charging behavior of electric vehicle in a single day and road congestion.

increases, and the charging demand of EVs decreases, while the energy price is at the lowest value. At 13:00, the output of photovoltaic power generation is the largest, and the demand for charging EVs is gradually increasing, with the highest price in one day. During the 16:00~18:00, the road congestion degree suddenly increased to above 0.7 due to getting off work.

Figs. 14 and 15, the histogram denotes the amount of electricity of the EV, and the three shaded histograms denotes the amount of electricity charged by the EV in the charging station, and the triangular curve in Fig. 14 denotes energy price, the square curve in Fig. 15 indicates the degree of road congestion at different times of the day, and the dotted line indicates the minimum amount of electricity required for an EV to reach the charging station. From Figs. 14 and 15, we can see that from 0:00 to 13:00, the power of EVs gradually decreases close to the threshold. Therefore, the EV must choose a charging station for charging. Through our EVSAC algorithm, EV finally chooses the charging at 14:00. Although

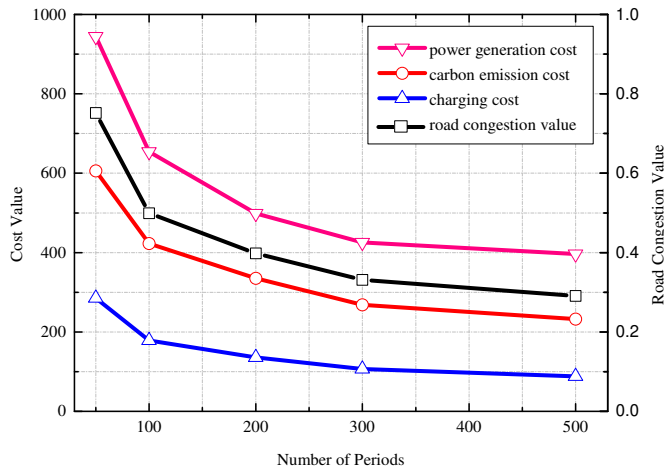


Fig. 16. The relationship between two-stage optimization objectives and iteration periods.

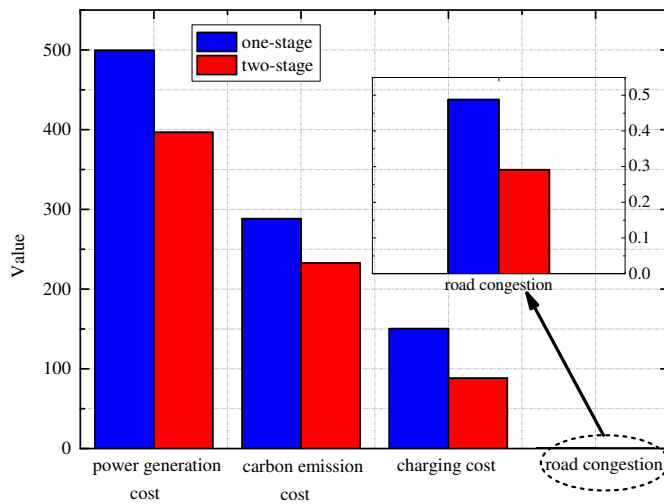


Fig. 17. The comparison between two-stage optimization objectives and one-stage optimization objectives.

the energy price at that moment is not the lowest price in the day, the congestion degree of EV on the road at this time is around 0.3, which is non-congestion. On the premise that EV can drive to the charging station at the highest speed, it can also charge at a lower energy price. Therefore, the EVSAC algorithm can offer a proper charging strategy for both EVs and the intelligent transportation systems.

Figs. 16 illustrates the relationship between the iteration periods and the system objectives in the two-stage optimization framework. As observed, with the increase in the number of iterations, the power generation cost, carbon emissions cost, charging cost, and road congestion value all exhibit a declining trend. This demonstrates that the two stages iteratively adjust to each other through multiple rounds of alternation, thereby enabling coordinated system-level optimization. A greater number of iteration periods leads to improved coordination among the objectives and more precise optimization outcomes,

further validating the effectiveness of the two-stage optimization approach in addressing multi-objective coordination.

Figs. 17 shows the comparison between the two-stage optimization approach and the one-stage optimization method. As shown in the figure, directly solving the combined optimization problem in a one-stage results in higher values across all four optimization objectives, compared to the two-stage sequential optimization. This outcome highlights the complexity of the problem, which involves multiple objectives, numerous constraints, and integer variables, making it an NP-hard problem. Consequently, the single-stage model suffers from poor convergence and requires considerable computational time to reach acceptable solutions. To overcome these challenges, this study proposes a two-stage optimization framework, incorporating DRL to improve both optimization efficiency and solution quality.

VI. CONCLUSION

In this paper, we designed a smart charging strategy for EVs by considering economic dispatch in cyber physical energy system. We have achieved more intelligent and efficient energy management by introducing mobile edge computing, collecting data by part of edge devices, and learning strategies in edge servers. To promote the accommodation of renewable energy, we designed the algorithm EDSAC to optimize renewable energy and traditional energy generation to dispatch a reasonable proportion of energy to each charging station. In addition, with the charging strategy offered by the EVSAC, EVs can select the most suitable charging station on their own charging needs. To address the complexity of joint energy and charging optimization, we developed a two-stage optimization framework. The first stage focuses on economic dispatch, while its outputs, such as electricity prices and grid loads, directly guide the EV charging decisions in the second stage. In turn, the aggregated charging behavior feeds back into the next round of dispatch optimization. Through iterative interactions, this framework enables system-level closed-loop coordination. Numerical results demonstrate that this two-stage method effectively reduces power generation costs, carbon emissions cost, EV charging costs, and road congestion across iterations, showing strong performance in multi-objective optimization.

REFERENCES

- [1] Q. Dong, D. Niyato, P. Wang, and Z. Han, "The phev charging scheduling and power supply optimization for charging stations," *IEEE Transactions on Vehicular Technology*, vol. 65, no. 2, pp. 566–580, Feb 2016.
- [2] W.-L. Liu, Y.-J. Gong, W.-N. Chen, Z. Liu, H. Wang, and J. Zhang, "Coordinated charging scheduling of electric vehicles: A mixed-variable differential evolution approach," *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, no. 12, pp. 5094–5109, Dec 2020.
- [3] M. Wang, M. Ismail, R. Zhang, X. Shen, E. Serpedin, and K. Qaraqe, "Spatio-temporal coordinated v2v energy swapping strategy for mobile pevs," *IEEE Transactions on Smart Grid*, vol. 9, no. 3, pp. 1566–1579, 2018.
- [4] T. Alladi, S. Chakravarty, V. Chamola, and M. Guizani, "A lightweight authentication and attestation scheme for in-transit vehicles in iov scenario," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 12, pp. 14 188–14 197, Dec 2020.
- [5] R. H. AlNahhal, A. F. Naiem, M. F. Shaaban, and M. Ismail, "Optimal planning of parking lots of pevs incorporating v2g for reliability improvement of distribution systems," *IEEE Access*, vol. 10, pp. 123 521–123 533, 2022.

- [6] Z. Fotouhi, M. R. Hashemi, H. Narimani, and I. S. Bayram, "A general model for ev drivers' charging behavior," *IEEE Transactions on Vehicular Technology*, vol. 68, no. 8, pp. 7368–7382, Aug 2019.
- [7] M. F. Shaaban, M. Ismail, E. F. El-Saadany, and W. Zhuang, "Real-time pev charging/discharging coordination in smart distribution systems," *IEEE Transactions on Smart Grid*, vol. 5, no. 4, pp. 1797–1807, 2014.
- [8] F. Zhang, Q. Yang, and D. An, "Cddpg: A deep-reinforcement-learning-based approach for electric vehicle charging control," *IEEE Internet of Things Journal*, vol. 8, no. 5, pp. 3075–3087, March 2021.
- [9] Q. Guo, S. Xin, H. Sun, Z. Li, and B. Zhang, "Rapid-charging navigation of electric vehicles based on real-time power systems and traffic data," *IEEE Transactions on Smart Grid*, vol. 5, no. 4, pp. 1969–1979, July 2014.
- [10] Q. Tang, K. Wang, Y. Song, F. Li, and J. H. Park, "Waiting time minimized charging and discharging strategy based on mobile edge computing supported by software-defined network," *IEEE Internet of Things Journal*, vol. 7, no. 7, pp. 6088–6101, July 2020.
- [11] M. Shin, D.-H. Choi, and J. Kim, "Cooperative management for pv/ess-enabled electric vehicle charging stations: A multiagent deep reinforcement learning approach," *IEEE Transactions on Industrial Informatics*, vol. 16, no. 5, pp. 3493–3503, May 2020.
- [12] T. Zhang, W. Chen, Z. Han, and Z. Cao, "Charging scheduling of electric vehicles with local renewable energy under uncertain electric vehicle arrival and grid power price," *IEEE Transactions on Vehicular Technology*, vol. 63, no. 6, pp. 2600–2612, July 2014.
- [13] X. Li, Y. Xiang, L. Lyu, C. Ji, Q. Zhang, F. Teng, and Y. Liu, "Price incentive-based charging navigation strategy for electric vehicles," *IEEE Transactions on Industry Applications*, vol. 56, no. 5, pp. 5762–5774, Sept-Oct 2020.
- [14] K. E. Adetunji, I. W. Hofsajer, A. M. Abu-Mahfouz, and L. Cheng, "A two-tailed pricing scheme for optimal ev charging scheduling using multiobjective reinforcement learning," *IEEE Transactions on Industrial Informatics*, vol. 20, no. 3, pp. 3361–3370, 2024.
- [15] C. Shao, K. Li, X. Li, Z. Hu, M. Shahidehpour, and X. Wang, "A decentralized bi-level decomposition method for optimal operation of electric vehicles in coupled urban transportation and power distribution systems," *IEEE Transactions on Transportation Electrification*, vol. 10, no. 1, pp. 2235–2246, 2024.
- [16] Y. Zhang, S. Dey, and Y. Shi, "Optimal vehicle charging in bilevel power-traffic networks via charging demand function," *IEEE Transactions on Smart Grid*, vol. 15, no. 3, pp. 3054–3065, 2024.
- [17] S. Su, Y. Li, K. Yamashita, M. Xia, N. Li, and K. A. Folly, "Electric vehicle charging guidance strategy considering "traffic network-charging station-driver" modeling: A multiagent deep reinforcement learning-based approach," *IEEE Transactions on Transportation Electrification*, vol. 10, no. 3, pp. 4653–4666, 2024.
- [18] X. Yang, T. Cui, H. Wang, and Y. Ye, "Multiagent deep reinforcement learning for electric vehicle fast charging station pricing game in electricity-transportation nexus," *IEEE Transactions on Industrial Informatics*, vol. 20, no. 4, pp. 6345–6355, 2024.
- [19] Y. Cao, J. Yi, Y. Liu, C. Zhao, D. Li, Y. Zhang, and Z. Han, "Joint routing and charging optimization of electric passenger vehicles with uninterruptible charging service," *IEEE Internet of Things Journal*, vol. 11, no. 10, pp. 18 180–18 192, 2024.
- [20] A. Habbal and M. F. Alrifai, "A user-preference-based charging station recommendation for electric vehicles," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 9, pp. 11 617–11 634, 2024.
- [21] L. Wang, L. Hou, S. Liu, Z. Han, and J. Wu, "Reinforcement contract design for vehicular-edge computing scheduling and energy trading via deep q-network with hybrid action space," *IEEE Transactions on Mobile Computing*, pp. 1–14, Nov 2023.
- [22] Y. Liang, Z. Ding, T. Ding, and W.-J. Lee, "Mobility-aware charging scheduling for shared on-demand electric vehicle fleet using deep reinforcement learning," *IEEE Transactions on Smart Grid*, vol. 12, no. 2, pp. 1380–1393, March 2021.
- [23] H. Li, Z. Wan, and H. He, "Constrained ev charging scheduling based on safe deep reinforcement learning," *IEEE Transactions on Smart Grid*, vol. 11, no. 3, pp. 2427–2439, May 2020.
- [24] A. A. Zishan, M. M. Haji, and O. Ardakanian, "Adaptive congestion control for electric vehicle charging in the smart grid," *IEEE Transactions on Smart Grid*, vol. 12, no. 3, pp. 2439–2449, May 2021.
- [25] T. Long, Q.-S. Jia, G. Wang, and Y. Yang, "Efficient real-time ev charging scheduling via ordinal optimization," *IEEE Transactions on Smart Grid*, vol. 12, no. 5, pp. 4029–4038, Sept 2021.
- [26] Y. Yu, C. Su, X. Tang, B. Kim, T. Song, and Z. Han, "Hierarchical game for networked electric vehicle public charging under time-based billing model," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 1, pp. 518–530, Jan 2021.
- [27] D. Yang, X. Zhou, Z. Yang, Y. Guo, and Q. Niu, "Low carbon multi-objective unit commitment integrating renewable generations," *IEEE Access*, vol. 8, pp. 207 768–207 778, 2020.
- [28] V. Sarfi and H. Livani, "An economic-reliability security-constrained optimal dispatch for microgrids," *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6777–6786, Nov 2018.
- [29] X. Lu, K. W. Chan, S. Xia, B. Zhou, and X. Luo, "Security-constrained multiperiod economic dispatch with renewable energy utilizing distributionally robust optimization," *IEEE Transactions on Sustainable Energy*, vol. 10, no. 2, pp. 768–779, April 2019.
- [30] D. Fang, X. Guan, B. Hu, Y. Peng, M. Chen, and K. Hwang, "Deep reinforcement learning for scenario-based robust economic dispatch strategy in internet of energy," *IEEE Internet of Things Journal*, vol. 8, no. 12, pp. 9654–9663, June 2021.
- [31] F. Jiao, Y. Zou, X. Zhang, and B. Zhang, "A three-stage multitime-scale framework for online dispatch in a microgrid with evs and renewable energy," *IEEE Transactions on Transportation Electrification*, vol. 8, no. 1, pp. 442–454, 2022.
- [32] K. Qu, Y. Chen, S. Xie, X. Zheng, and J. Zhu, "Segmented distributionally robust optimization for real-time power dispatch with wind uncertainty," *IEEE Transactions on Power Systems*, vol. 39, no. 2, pp. 2970–2983, 2024.
- [33] C. Mu, Y. Shi, N. Xu, X. Wang, Z. Tang, H. Jia, and H. Geng, "Multi-objective interval optimization dispatch of microgrid via deep reinforcement learning," *IEEE Transactions on Smart Grid*, vol. 15, no. 3, pp. 2957–2970, 2024.
- [34] Y. Xia, Y. Xu, and X. Feng, "Hierarchical coordination of networked-microgrids toward decentralized operation: A safe deep reinforcement learning method," *IEEE Transactions on Sustainable Energy*, vol. 15, no. 3, pp. 1981–1993, 2024.
- [35] Q. Wang, X. Zhang, Y. Xu, Z. Yi, and D. Xu, "A low-carbon economic dispatch method for power systems with carbon capture plants based on safe reinforcement learning," *IEEE Transactions on Industrial Informatics*, vol. 20, no. 8, pp. 10 542–10 553, 2024.
- [36] M. Carrion and J. Arroyo, "A computationally efficient mixed-integer linear formulation for the thermal unit commitment problem," *IEEE Transactions on Power Systems*, vol. 21, no. 3, pp. 1371–1378, 2006.
- [37] L. Lin, X. Guan, Y. Peng, N. Wang, S. Maharjan, and T. Ohtsuki, "Deep reinforcement learning for economic dispatch of virtual power plant in internet of energy," *IEEE Internet of Things Journal*, vol. 7, no. 7, pp. 6288–6301, July 2020.
- [38] X. Yang, H. He, Y. Zhang, Y. Chen, and G. Weng, "Interactive energy management for enhancing power balances in multi-microgrids," *IEEE Transactions on Smart Grid*, vol. 10, no. 6, pp. 6055–6069, 2019.
- [39] Z. Deng, D. Huang, J. Liu, B. Mi, and Y. Liu, "An assessment method for traffic state vulnerability based on a cloud model for urban road network traffic systems," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 11, pp. 7155–7168, Nov 2021.
- [40] M. R. Garey and D. S. Johnson, "Computers and intractability: A guide to the theory of np-completeness," *W. H. Freeman*, 1983.
- [41] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *International conference on machine learning*. PMLR, 2018, pp. 1861–1870.
- [42] M. Gong, Y. Zhao, J. Sun, C. Han, G. Sun, and B. Yan, "Load forecasting of district heating system based on informer," *Energy*, vol. 253, p. 124179, 2022.
- [43] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informer: Beyond efficient transformer for long sequence time-series forecasting," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 35, no. 12, 2021, pp. 11 106–11 115.
- [44] Y. Chen, J. Ding, and Q. Chen, "A reinforcement learning based large-scale refinery production scheduling algorithm," *IEEE Transactions on Automation Science and Engineering*, vol. 21, no. 4, pp. 6041–6055, 2024.
- [45] C. Qiu, Y. Hu, Y. Chen, and B. Zeng, "Deep deterministic policy gradient (ddpg)-based energy harvesting wireless communications," *IEEE Internet of Things Journal*, vol. 6, no. 5, pp. 8577–8588, Oct 2019.
- [46] "Epic (elia portal interface for customer)," [OnLine], accessed August 02, 2021. <https://www.elia.be/en/grid-data/power-generation/>.
- [47] T. Qian, C. Shao, X. Wang, and M. Shahidehpour, "Deep reinforcement learning for ev charging navigation by coordinating smart grid and intelligent transportation system," *IEEE Transactions on Smart Grid*, vol. 11, no. 2, pp. 1714–1723, March 2020.